

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

January 2025

Grade 12 Pre-Calculus Mathematics Achievement Test:
Booklet 2 (January 2025)

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Available in alternate formats upon request.

Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time Required to Complete the Test: 3 hours

Additional Time Allowed: 30 minutes

Number and Marks by Question Type

	Selected Response	Constructed Response	Marks
Booklet 1	—	16	40
Booklet 2	9	19	50
Total	9	35	90

Note that diagrams and graphs provided in the test booklets may not be drawn to scale.

DIRECTIONS

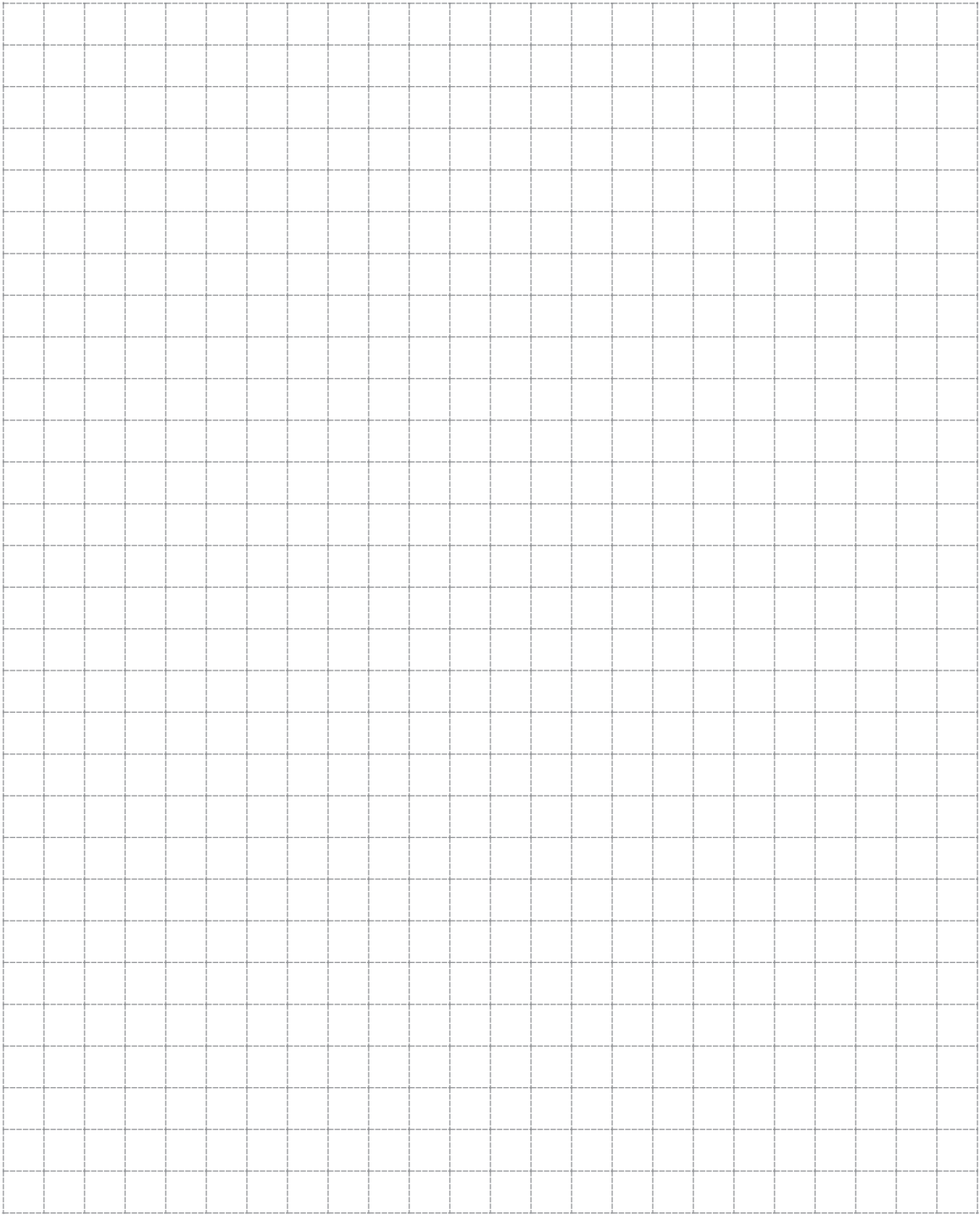
Selected Response Questions

- Calculators are **not** allowed for this part of the test.
- You may use the spaces beside each question for rough work.
- Provide only one answer per question.
- There is no penalty for guessing.
- Record your answers on the sheet provided.

Constructed Response Questions

- Calculators are **not** allowed for this part of the test.
- Write each solution in the space provided.
- For full marks, your answer must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, clear, and well organized.

NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.



Question 17**1 mark**

Identify the function that has a different range than $f(x) = x^2$.

- a. $y = 4f(x)$
- b. $y = f(x) + 4$
- c. $y = f(x + 4)$
- d. $y = f(-x)$

Question 18**1 mark**

Identify the exact value of $\sin\left(-\frac{7\pi}{3}\right)$.

- a. $-\frac{\sqrt{3}}{2}$
- b. $-\frac{1}{2}$
- c. $\frac{1}{2}$
- d. $\frac{\sqrt{3}}{2}$

Question 19**1 mark**

Identify the value of a in the equation, $\log_a 3 = \frac{1}{4}$.

- a. $\frac{1}{64}$
- b. $3^{\frac{1}{4}}$
- c. 12
- d. 81

Question 20**1 mark**

Identify a value of x where the graphs of $y = \cos x$ and $y = \sin x$ intersect.

a. $-\frac{3\pi}{4}$

b. $-\frac{\pi}{4}$

c. 0

d. $\frac{3\pi}{4}$

Question 21**1 mark**

Identify the number of negative terms in the binomial expansion of $(2x - y)^6$.

a. 0

b. 3

c. 4

d. 6

Question 22**1 mark**

Identify the general solution of the equation, $\sin x = 1$.

a. $x = k\pi, k \in \mathbb{Z}$

b. $x = 2k\pi, k \in \mathbb{Z}$

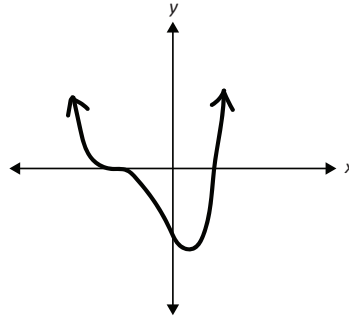
c. $x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$

d. $x = \frac{\pi}{2} + 2k\pi, k \in \mathbb{Z}$

Question 23**1 mark**

Identify a possible degree of the polynomial function represented below.

- a. 2
- b. 3
- c. 4
- d. 5

**Question 24****1 mark**

Identify the expression that is equivalent to $\cos x \cot x$.

- a. $\frac{1 - \sin x}{\sin x}$
- b. $\frac{\sin^2 x - 1}{\sin x}$
- c. $\sin x$
- d. $\frac{1 - \sin^2 x}{\sin x}$

Question 25**1 mark**

Given that the point $(-7, -5)$ is on the graph of $y = g(x)$, identify the corresponding point on the graph of $y = |g(x)|$.

- a. $(7, 5)$
- b. $(-7, 5)$
- c. $(7, -5)$
- d. $(-5, -7)$

Question 26**2 marks**

118

Given that $x = 1$ is one of the zeros of $p(x) = x^3 + x^2 - 10x + 8$, express $p(x)$ in completely factored form.

$p(x) =$ _____

Question 27**2 marks**

119

Determine the exact value of $\frac{\csc^2 \theta - 1}{1 - \cos^2 \theta}$ if $\theta = \frac{5\pi}{6}$.

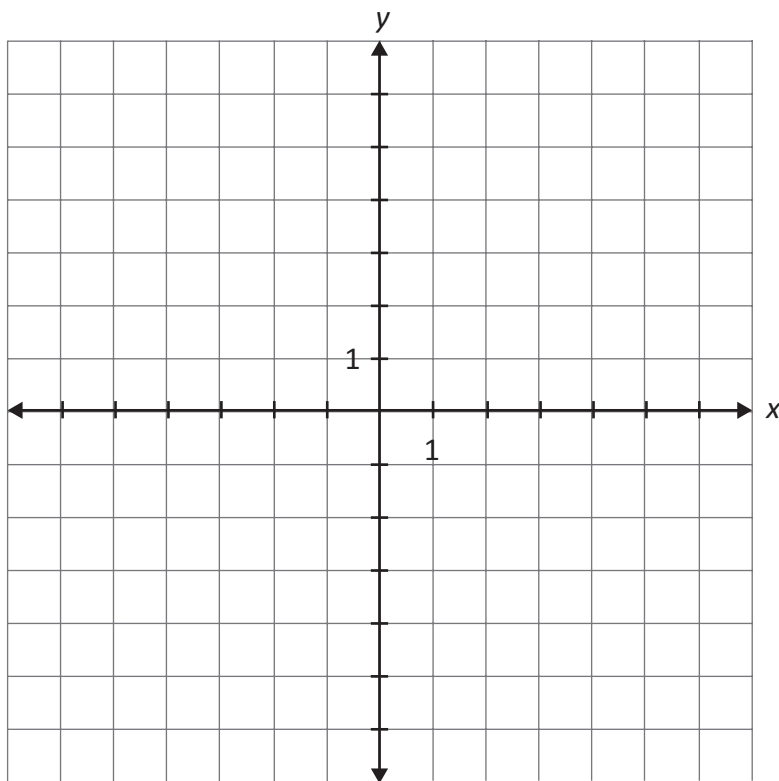
Question 28

3 marks

120

Solve, graphically.

$$-2x + 5 = \sqrt{x - 1}$$



Question 29

2 marks

121

If $\log_a 2 = p$ and $\log_a 5 = q$, express $\log_a 40$ in terms of p and q .

Question 30

1 mark

122

Describe how to sketch the graph of $y=f^{-1}(x)$, given the graph of a linear function, $y=f(x)$.

Question 31**3 marks**

123

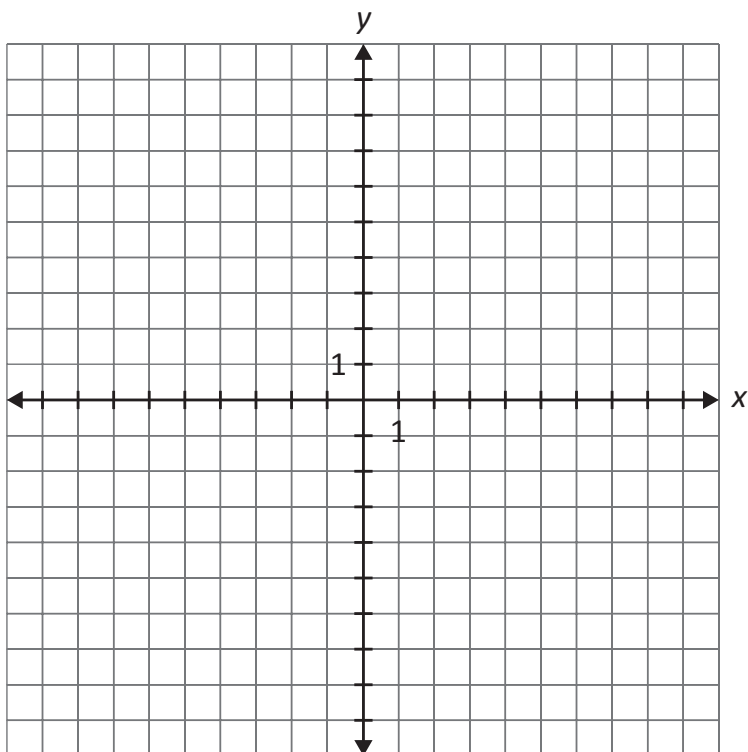
Given that $\sin \alpha = -\frac{4}{5}$ where α is in quadrant III, and $\cos \beta = \frac{12}{13}$ where β is in quadrant IV, determine the exact value of $\cos(\alpha - \beta)$.

Question 32

3 marks

124

Sketch the graph of $f(x) = \frac{2x-4}{x+1}$.



Question 33

a) 1 mark b) 1 mark c) 1 mark

125
126
127

Due to the tidal variations, the depth of the water in a bay is represented by the function,

$$d(t) = -2\cos\left[\frac{\pi}{6}(t+4)\right] + 16$$

where d is the depth of the water, in metres and
 t is the time, in hours.

a) Determine the depth of the water when $t = 2$.

b) State the minimum depth of the water.

c) Determine the length of time between two consecutive low tides (minimum depths).

Question 34

3 marks

128

Evaluate.

$$\sin^2\left(\frac{5\pi}{4}\right)\sec\left(\frac{13\pi}{3}\right) - \cot\left(\frac{7\pi}{4}\right)$$

Question 35

a) 1 mark b) 1 mark

129
130

a) State the non-permissible value(s) of the function, $f(x) = \frac{x-3}{x^2-9}$.

b) Describe what each non-permissible value represents on the graph of $f(x) = \frac{x-3}{x^2-9}$.

Question 36

1 mark

131

Determine which of the given values is greater. Justify your answer.

$$\log_3 38 \text{ or } \log_4 50$$

Question 37**1 mark**

132

Determine if the point $\left(\frac{3}{4}, \frac{\sqrt{5}}{4}\right)$ is on the unit circle. Justify your answer.

Question 38**2 marks**

133

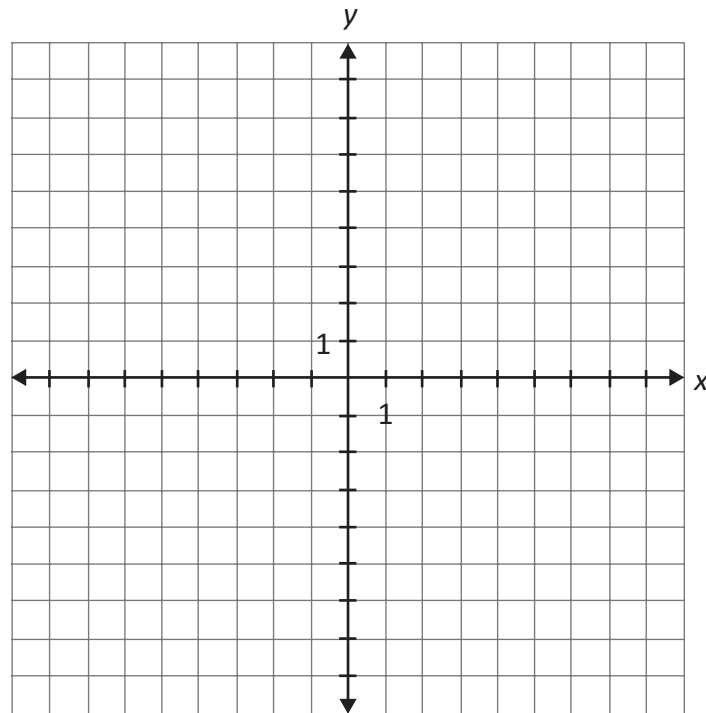
Determine which term contains x^5 in the binomial expansion of $\left(2x + \frac{3}{x}\right)^{11}$.

Question 39

3 marks

134

Sketch the graph of $y = 2(3^x) + 1$.



Question 40

2 marks

135

Solve, algebraically.

$$\log(2x + 1) = \log(x - 2) + 1$$

Question 41

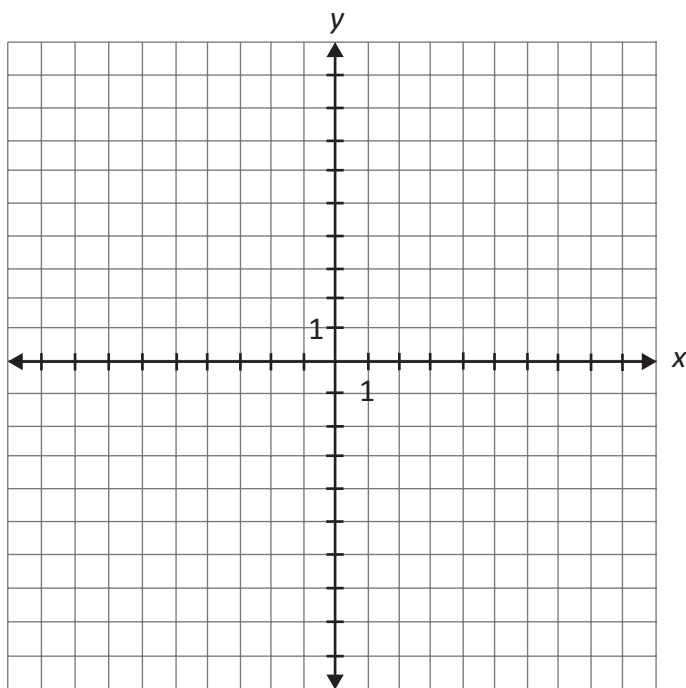
a) 1 mark b) 1 mark

136
137

Given the functions, $f(x) = x + 2$ and $g(x) = x - 2$,

a) state the equation of $h(x) = f(x) \cdot g(x)$.

b) sketch the graph of $h(x) = f(x) \cdot g(x)$.



Question 42**1 mark**

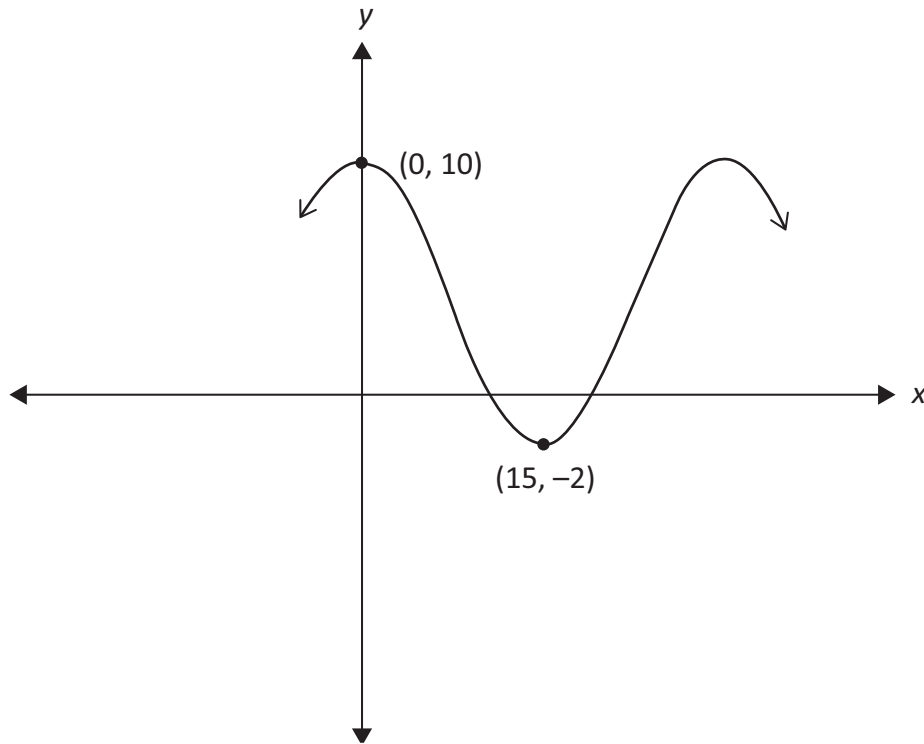
138

Given $p(x) = -2x^3 + 3x + 1$, determine the remainder when $p(x)$ is divided by $(x - 2)$.

Question 43**3 marks**

139

Determine the equation of the sinusoidal function in the form $y = A\cos(Bx) + D$.



$y =$ _____

Question 44**2 marks**

140

Determine the total number of arrangements of the letters in the word TORONTO.

State your answer as a whole number.